The title (*which should really have more than two words, but should not be longer than two lines*)

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Abstract

TODO

1 Introduction

The Domain Name System (DNS) is a fundamental component of the Internet, responsible for translating human-readable domain names into IP addresses (Mockapetris, 1987). Most interactions with the internet require DNS resolution, which makes DNS queries a source of information for users' online activity. Originally, DNS queries have been transmitted without encryption, allowing all intermediaries with access to the packet to observe which domains users access (Wicinski, 2021). Protocols such as DNS over HTTPS (DoH), DNS over TLS (DoT) and DNS over QUIC (DoQ) have been invented to provide a solution for this issue by encrypting the communication between clients and DNS servers (Hoffman and McManus, 2018; Hu et al., 2016; Huitema et al., 2022). However, while these protocols protect DNS queries from intermediaries in the network, the DNS server can still associate a query to the IP Address of the user that issued them.

Oblivious DNS over HTTPS (ODoH) addresses this issue by introducing an intermediary proxy that separates the client's identity from the query contents, forwarding encrypted requests to the target DNS server, which ensures that neither the proxy or the DNS server possess both the user's identity and the query contents (Singanamalla et al., 2020; Kinnear et al., 2022). Although this approach guarantees the privacy of the client's request end to end, it also prevents intermediary proxies from caching requests, since the query contents only get decrypted at the target resolver.

To overcome this issue, Niroula et al. (2026) developed Cacheable Oblivious DNS over HTTPS (CODoH) protocol. Rather than simply decrypting and caching DNS queries, CODoH combines trusted execution environments with techniques to prevent cache access patterns from revealing information. Its objective is to preserve the privacy properties of ODoH while partly recovering the performance benefits of DNS caching. However, these mechanisms also introduce overhead, raising questions about the performance of CODoH. This report examines the architecture and privacy-enhancing mechanisms of CODoH.

2 Background

2.1 DNS & DNS Caching

The *Domain Name System (DNS)* is a distributed naming system that allows mapping domain names into certain information, usually into IP addresses (Mockapetris, 1987). When a client needs to resolve a domain address, usually it sends a query to a recursive *DNS resolver*, which may contact multiple DNS servers following the DNS hierarchy, as illustrated in Figure 1. For example, a client may request the DNS resolver the resolution of the domain "example.com". In such case, the DNS resolver will initially query the Root Server to get the Top Level Domain (TLD) server information. Then, it will query the TLD server ".com" to obtain the authoritative Name Server (NS) of "example.com". Finally, it will query the NS to obtain the corresponding IP Address, and answer the original client.

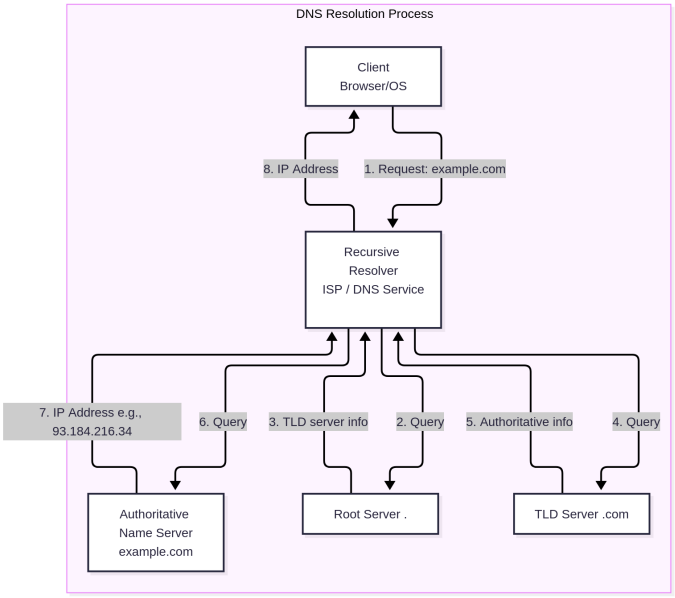


Figure 1: DNS resolution process. Initially generated using Gemini 4 Argon with the prompt "Generate a high-level mermaid diagram illustrating DNS resolution with a recursive resolver" on 17.10.2026, modified by the author.

To reduce the overhead associated with the DNS resolution process, recursive resolvers may maintain caches of previously retrieved DNS records, with records having an associated Time To Live (TTL), which determines for how long the DNS server can deliver the stored information until it is considered stale and discarded. (Mockapetris, 1987) When a resolver receives a query, it checks whether there is a valid cached answer, which would result in a *cache hit*. Otherwise, a *cache miss* happens, which forces the resolve to query the authoritative servers again. This process is illustrated in Figure 2

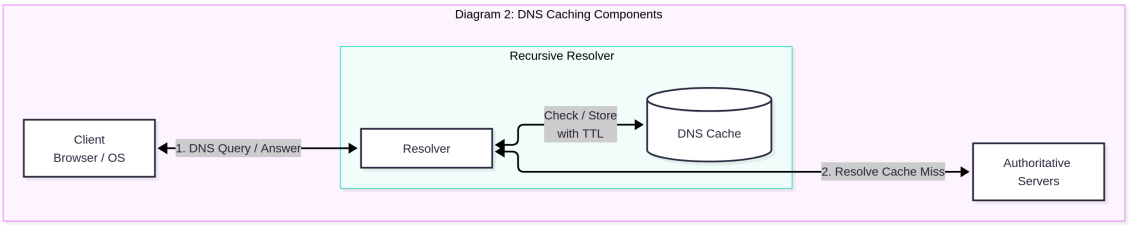


Figure 2: DNS caching. Initially generated using Gemini 4 Argon with the prompt "Generate a high-level mermaid diagram illustrating DNS recursive resolver caching, with its components and interactions" on 17.10.2026, modified by the author.

2.2 DNS Privacy: DoH & ODoH

Traditionally, DNS messages have been transported without encryption, usually encapsulated over User Datagram Protocol (UDP) (Mockapetris, 1987). This implies that network observers can read all the packet information, including Source IP Address and the requested domain name, potentially exposing browsing activity. Furthermore, the lack of encryption in traditional DNS increases its vulnerability to network attacks. An attacker may intercept a query and try to manipulate the response to redirect the victim to a malicious location (Atkins and Austein, 2004).

Hoffman and McManus (2018) solved the lack of encryption by transporting DNS over HTTPS, using Transport Layer Security (TLS) for encryption, integrity and confidentiality. This is done by encapsulating DNS messages in HTTP requests under HTTPS URIs, using either POST or GET methods. DoH prevents network observers and/or attackers from reading the DNS queries, partly protecting users' privacy. However, it still has a key limitation: the resolver has access to the unencrypted DNS message and can associate

it with the client's IP address.

To solve this problem, Singanamalla et al. (2020) introduced *Oblivious DNS over HTTPS (ODoH)*, which introduces a proxy between the client and the resolver. The client encrypts the DNS message for the resolver, so the proxy does not have access to the contents of the query, and forwards the encrypted DNS message, as illustrated in Figure 3. Assuming the proxy and the resolver do not collude, the client's IP address is unknown for the resolver, and the queried domain name is unknown for the proxy, so the users' privacy remains protected. However, this introduces a new problem: encryption makes two equal DNS queries different to the proxy, so conventional caching becomes impossible (Niroula et al., 2026).

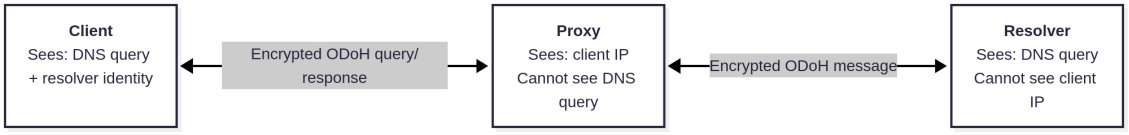


Figure 3: ODoH Architecture. Generated using GPT 5.6 Sol High with the prompt "Create a simple mermaid LR diagram to illustrate ODoH: Client <-> Proxy <-> Resolver. Show what's visible to each party" on 17.10.2026

2.3 Trusted Execution Environments (TEEs) & Remote Attestation

Although encryption protects sensitive information in transit and rest, processing this information usually requires it to be decrypted and loaded into memory. Traditionally, privileged software such as the Operating System has been allowed to access and manipulate all computer memory. Consequently, applications must trust the environment in which they execute. *Trusted Execution Environments (TEEs)* address this limitation by providing hardware-enforced isolation. (Sabt et al., 2015)

Intel SGX (Costan and Devadas, 2016) is a TEE that extends the Intel architecture to guarantee isolated execution in protected containers called *enclaves*, and permits proving to a remote computer that it is communicating with a specific secure container using *software attestation*. The proof is an attestation signature covering the container's initial state, a challenge random number produced by the remote computer, and a message produced by the container. This allows remote hosts to execute programs in a secure container, as illustrated in Figure 5.

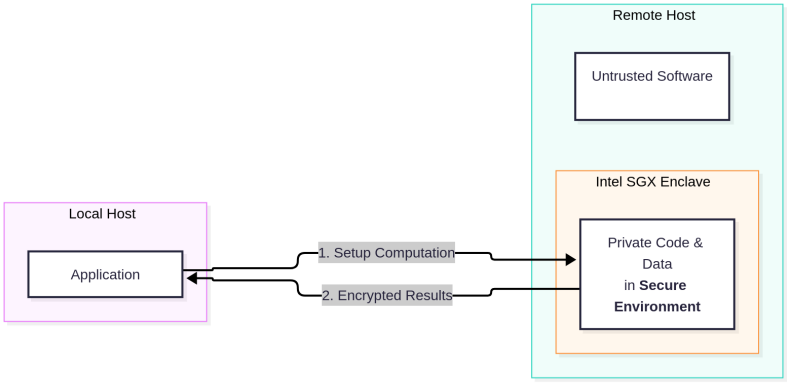


Figure 4: Intel SGX Secure Remote Computation. A local host setups computation in a remote secure container (enclave), and receives encrypted results from that computation.

2.4 Oblivious RAM (ORAM)

Even though Intel SGX's enclaves have their memory encrypted (Costan and Devadas, 2016), this alone may not be enough to protect their data. An attacker observing memory access patterns may be able to infer sensitive information, even if the underlying information is encrypted. Goldreich (1987) introduced *Oblivious RAM (ORAM)* to address this problem by transforming how memory is accessed, preventing observable access patterns from revealing what data is being accessed.

Path ORAM (Stefanov and Shi, 2012) is a modern ORAM protocol that organizes encrypted data blocks by assigning them to leafs in a binary tree. After each access, the block is moved to a different leaf, preventing repeated access to the same data block from revealing access patterns. However, this introduces significant memory access overhead when compared to traditional memory access.

2.5 Related work

Here, you include short descriptions of similar work and explain *how* they are different from the main content you have presented.

Level of detail similar to an abstract.

3 CODoH Design

3.1 Design Goals

ODoH separates the knowledge between two parties: the proxy, aware of the user's IP address, and the target, with access to the unencrypted DNS query (Singanamalla et al., 2020). However, effective conventional DoH proxy caching requires the proxy to be able to identify repeated requests (Hoffman and McManus, 2018), which is not possible with CODoH's encryption. CODoH's fundamental goal therefore is to implement a cache at the proxy while preserving ODoH's privacy enhancements (Niroula et al., 2026), in order to achieve *performance requirements* and *cache-specific security goals*.

Performance requirements. CODoH should reduce latency for cache hits relative to ODoH baseline; for cache misses, latency should not significantly differ relative to ODoH baseline. (Niroula et al., 2026)

Cache-specific attack vectors. Caching introduces new attack vectors (Niroula et al., 2026). For example, in a *set-difference (bracketing) attack*, a malicious proxy may:

(1) Learn the current cache state.

(2) Allow a victim query to proceed, queueing other traffic.

(3) Learn the cache state again.

(4) Compute the difference to isolate the newly added entry, potentially identifying the victim's query.

Cache-specific security goals. Given the new attack vectors, CODoH aims to achieve the security goals below:

(G1) *Authorized and fresh cached answer*. A proxy must not be able to cause clients to accept answers not authorized by the target, or answers with an expired TTL.

(G2) *Cache-state indistinguishability*. The proxy must not be able to learn meaningful information about particular queries by comparing cache states across time, like in the previous example of a set-difference attack.

(G3) *Query equality and profiling resistance*. The proxy must not be able to determine whether two DNS queries are equal by comparing them.

3.2 Design Overview

CODoH (Niroula et al., 2026) aims to achieve its goals by following an architecture, illustrated in Figure 5, with three independent parties: the *client*, the *proxy*, which contains the *enclave*, and the *DNS resolver*

The client is expected to have access to the HPKE public keys of the enclave and the target. Using these, it encrypts the query for both the enclave (CODoH query) and the target (ODoH query), and sends both encrypted queries to the proxy. Finally, it receives both encrypted DNS responses from the proxy, and decrypts them.

The proxy forwards both encrypted messages to the enclave and the target. In addition, it forwards encrypted responses from the enclave and the target, whose contents are encrypted, so whether it is a cache hit or miss is unknown to the proxy, and the client always needs to receive both responses.

The enclave contains the cache table, accessed using Path ORAM to avoid access patterns leaking information. Cache gets filled with signed and encrypted response batches received from the target, which include a trusted timestamp that ensures safe cache expiration and replay protection.

The target decrypts the DNS query and answers with an encrypted response whose private key is only known by the client. It has no way to identify the client, so privacy is preserved.

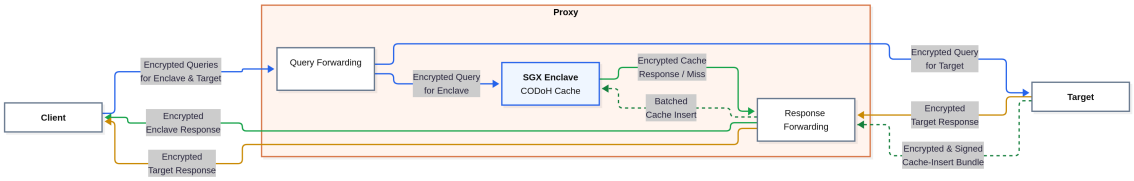


Figure 5: Simplified CODoH Architecture, adapted from Niroula et al. (2026). The proxy forwards independently encrypted queries to the SGX enclave and target. The enclave provides privacy-preserving cache. Target provides signed and encrypted cache insert bundles with a trusted timestamp. Solid arrows represent queries and responses, while dashed arrows represent cache inserts. Blue arrows represent queries, red arrows represent cached responses and cache inserts, and yellow arrows represent target responses.

3.3 Security & Privacy

3.4 CODoH's Limitations

3.5 Performance

4 Another topic in the main content

5 Conclusion

6 Use of generative digital tools

Large Language Models (LLMs), specifically Gemini and ChatGPT models, have been used in the creation of this report for the following purposes:

A. Research and understanding concepts, using prompts of the type: "How does [...] work?", "Explain [...]", "What solutions similar to [...] have been used to solve [...] problem?", "What is [...]?", "What problem does [...] solve?"

B. Generation of diagrams. All prompts were of the type: "Generate a mermaid diagram illustrating [...]"

C. Concept checking. Prompts of the type: "Is [...] conceptually correct?", "Is [...] correctly expressed?"

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